

SESSION-1

Task 1: To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Make file.

CODE:

```
UW PICO 5.09 File: 13.c

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>

int main()
{
    printf("=====\\n");
    printf("      LINUX OS PROJECT SETUP\\n");
    printf("=====\\n\\n");

    /* Operating System Information */
    printf("1. Operating System Information\\n");
    printf("-----\\n");
    system("uname -a");

    /* GCC Information */
    printf("2. GCC Compiler Information\\n");
    printf("-----\\n");
    system("gcc --version");

    /* Shell Information */
    printf("3. Shell Information\\n");
    printf("-----\\n");
    printf("Shell: %s\\n", getenv("SHELL"));

    /* Process Information */
    printf("4. Process Information\\n");
    printf("-----\\n");
    printf("Process ID (PID) : %d\\n", getpid());
    printf("Parent PID (PPID) : %d\\n", getppid());

    /* Git Information */
    printf("5. Git Information\\n");
    printf("-----\\n");
    system("git --version");

    /* Project Structure */
    printf("6. Project Structure\\n");
    printf("-----\\n");
    system("ls -la");

    printf("=====\\n");
    printf("      SETUP CHECK COMPLETED\\n");
    printf("=====\\n");

    return 0;
}
```

OUTPUT:

```
-rw-r--r-- 1 nrk staff 537 25 Aug 11:47 .zprofile
-rw-r--r-- 1 nrk staff 166 15 Jun 08:06 .zprofile.py3save
-rw-r--r-- 1 nrk staff 3989 29 Aug 08:39 .zsh_history
drwx----- 58 nrk staff 1680 29 Aug 08:46 .zsh_sessions
-rw-r--r-- 1 nrk staff 1447 26 Aug 12:06 l.c.save
-rwxr-xr-x 1 nrk staff 33472 29 Aug 08:36 11
-rw-r--r-- 1 nrk staff 802 29 Aug 08:35 11.c
-rwxr-xr-x 1 nrk staff 33600 29 Aug 08:47 13
-rw-r--r-- 1 nrk staff 1479 29 Aug 08:47 13.c
-rwxr-xr-x 1 nrk staff 33584 26 Aug 12:11 2
-rw-r--r-- 1 nrk staff 1990 26 Aug 12:11 2.c
-rwxr-xr-x 1 nrk staff 33936 26 Aug 12:15 3
-rw-r--r-- 1 nrk staff 1142 26 Aug 12:14 3.c
-rwxr-xr-x 1 nrk staff 33760 26 Aug 12:20 4
-rw-r--r-- 1 nrk staff 1517 26 Aug 12:26 4.c
-rwxr-xr-x 1 nrk staff 33968 26 Aug 12:04 444
-rw-r--r-- 1 nrk staff 1282 26 Aug 12:03 444.c
-rwxr-xr-x 1 nrk staff 33760 26 Aug 12:28 5
-rw-r--r-- 1 nrk staff 1018 26 Aug 12:28 5.c
-rw-r--r-- 1 nrk staff 1394 26 Aug 11:57 a.c.save
drwxr-xr-x@ 4 nrk staff 128 16 Jul 17:12 airline-lost-found
-rw-r--r-- 1 nrk staff 1279292 24 May 19:31 ancient.dress.fbx
drwx-----@ 4 nrk staff 128 22 Mar 2025 Applications
-rw-r--r-- 1 nrk staff 1393 26 Aug 11:50 b.c.save
drwxr-xr-x@ 7 nrk staff 224 9 Jul 15:21 blenderkit_data
-rw-r--r--@ 1 nrk staff 4836 4 Jun 08:22 class AVLTree (.java
-rw-r--r--@ 1 nrk staff 1835 4 Jun 10:41 class RankNode (.java
drwxrwxrwx 2 nrk staff 64 22 Jun 13:04 DaVinci Resolve Media
-rw-r--r--@ 1 nrk staff 7131 29 Nov 2025 ddca.circ
drwx-----+ 4 nrk staff 128 12 Aug 08:37 Desktop
drwx-----+ 32 nrk staff 1024 29 Aug 08:38 Documents
drwx-----@ 97 nrk staff 3104 29 Aug 08:44 Downloads
drwxr-xr-x@ 3 nrk staff 96 15 Jun 10:35 EXAMINATION SEATING ARRANGEMENT
-rw-r--r--@ 1 nrk staff 317 4 Feb 2026 fed.css .h
-rw-r--r-- 1 nrk staff 2 25 Aug 12:00 file.c
-rw-r--r-- 1 nrk staff 16 25 Aug 12:03 fl.c
-rw-r--r--@ 1 nrk staff 2593 4 Jun 10:21 import java.util.java
-rw-r--r-- 1 nrk staff 1450 29 Aug 08:30 k1.c
drwx-----@ 100 nrk staff 3280 28 Jul 09:09 Library
-rwxr-xr-x 1 nrk staff 33664 29 Aug 08:33 lk
-rw-r--r-- 1 nrk staff 1493 29 Aug 08:32 lk.c
-rw-r--r--@ 1 nrk staff 2215 3 Nov 2025 logism.circ
drwx----- 8 nrk staff 256 11 Jul 14:43 Movies
drwx-----+ 5 nrk staff 160 9 Jul 15:22 Music
-rw-r--r-- 1 nrk staff 1777 26 Aug 12:00 n.c
drwxr-xr-x@ 2 nrk staff 64 12 Apr 15:36 New Folder
-rw-r--r--@ 1 nrk staff 3929 3 Jun 12:34 package com.mediflow.java
drwx-----+ 5 nrk staff 160 21 Jul 11:34 Pictures
-rw-r--r--@ 1 nrk staff 917 23 Feb 2026 project.html
drwxr-xr-x+ 4 nrk staff 128 31 Jul 2024 Public
-rw-r--r--@ 1 nrk staff 2829 4 Jun 10:19 public class SegmentTreeLazy (.java
-rw-r--r-- 1 nrk staff 1777 26 Aug 11:50 r.c
drwxr-xr-x@ 5 nrk staff 160 15 Jun 10:17 rohith
-rw-r--r-- 1 nrk staff 1203 26 Aug 12:05 rohith.c
-rw-r--r-- 1 nrk staff 1393 26 Aug 11:50 space.save
-rw-r--r--@ 1 nrk staff 317 4 Feb 2026 student.html.h
-rw-r--r--@ 1 nrk staff 2604 16 Feb 2026 student.registration.form.html
-rw-r--r--@ 1 nrk staff 317 4 Feb 2026 student.html.h
-rw-r--r--@ 1 nrk staff 1804 3 Jun 12:32 Untitled-1.cs

=====
      SETUP CHECK COMPLETED
=====
nrk@nallas-mac-Book-Air ~ %
```

Task 2: To Analyse process abstraction, execute fork(), understand exec() family, analyse parent-child relationships, inspect process tree, practice system call tracing.

CODE:

```
UW PICO 5.09 File: 14.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    int status;

    printf("*****\n");
    printf("        PROCESS ABSTRACTION DEMO\n");
    printf("*****\n");

    printf("Parent Process Information\n");
    printf("-----\n");
    printf("Parent PID : %d\n", getpid());
    printf("Parent PPID : %d\n", getppid());
    printf("Process State: Running\n");

    /* Create child process */
    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

    if (pid == 0)
    {
        /* Child Process */
        printf("*****\n");
        printf("        CHILD PROCESS\n");
        printf("*****\n");

        printf("Child PID : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());
        printf("Process State: Running\n");

        printf("\nChild is going to execute 'ls' using exec().\n\n");

        /*
         * Replace child process with the ls program.
         * If execvp() succeeds, the code after it will not execute.
         */
        execvp("ls", "ls", "-l", NULL);

        /* This executes only if exec() fails */
        perror("exec failed");
        exit(1);
    }
    else
    {
        /* Parent Process */
        printf("*****\n");
        printf("        PARENT PROCESS\n");
        printf("*****\n");
    }
}
```

OUTPUT:

```
nrk@nallas-mac-Book-Air ~ % gcc 14.c -o 14
nrk@nallas-mac-Book-Air ~ % ./14
=====
PROCESS ABSTRACTION DEMO
=====

Parent Process Information
-----
Parent PID : 1986
Parent PPID : 1974
Process State: Running

=====
PARENT PROCESS
=====
Parent PID : 1986
Child PID : 1987
Parent PPID : 1974
Process State: Running

Parent is waiting for child...
Process State: Waiting
=====
CHILD PROCESS
=====
Child PID : 1987
Child PPID : 1986
Process State: Running

Child is going to execute 'ls' using exec().

total 3424
-rw----- 1 nrk staff 1447 26 Aug 12:06 1.c.save
-rwxr-xr-x 1 nrk staff 33472 29 Aug 08:36 11
-rw-r--r-- 1 nrk staff 802 29 Aug 08:35 11.c
-rwxr-xr-x 1 nrk staff 33608 29 Aug 08:47 13
-rw-r--r-- 1 nrk staff 1479 29 Aug 08:47 13.C
-rwxr-xr-x 1 nrk staff 33728 29 Aug 08:51 14
-rw-r--r-- 1 nrk staff 2349 29 Aug 08:51 14.c
-rwxr-xr-x 1 nrk staff 33804 26 Aug 12:11 2
-rw-r--r-- 1 nrk staff 1990 26 Aug 12:11 2.c
-rwxr-xr-x 1 nrk staff 33936 26 Aug 12:15 3
-rw-r--r-- 1 nrk staff 1142 26 Aug 12:14 3.c
-rwxr-xr-x 1 nrk staff 33768 26 Aug 12:28 4
-rw-r--r-- 1 nrk staff 1517 26 Aug 12:26 4.c
-rwxr-xr-x 1 nrk staff 33968 26 Aug 12:04 444
-rw-r--r-- 1 nrk staff 1202 25 Aug 12:03 444.c
-rwxr-xr-x 1 nrk staff 33768 26 Aug 12:28 5
-rw-r--r-- 1 nrk staff 1018 26 Aug 12:28 5.c
-rw----- 1 nrk staff 1394 26 Aug 11:57 a.c.save
drwxr-xr-x@ 4 nrk staff 128 16 Jul 17:12 airline-lost-found
-rw-r--r-- 1 nrk staff 1279292 24 May 19:31 ancient_dress.fbx
drwx-----@ 4 nrk staff 128 22 Mar 2025 Applications
-rw----- 1 nrk staff 1393 26 Aug 11:58 b.c.save
drwxr-xr-x@ 7 nrk staff 224 9 Jul 15:21 blenderkit_data
-rw-r--r--@ 1 nrk staff 4836 4 Jun 08:22 class AVLTree (.java
-rw-r--r--@ 1 nrk staff 1035 4 Jun 10:41 class RankNode (.java
drwxrwxrwx 2 nrk staff 64 22 Jun 13:04 DaVinci Resolve Media
-rw-r--r--@ 1 nrk staff 7131 29 Nov 2025 ddca.circ
drwx-----+ 4 nrk staff 128 12 Aug 08:37 Desktop
drwx-----+ 34 nrk staff 1088 29 Aug 08:48 Documents
drwx-----@ 97 nrk staff 3104 29 Aug 08:44 Downloads
drwxr-xr-x@ 3 nrk staff 96 15 Jun 10:35 EXAMINATION SEATING ARRANGEMENT
```

```
nrk@nallas-mac-Book-Air ~ % gcc 14.c -o 14
nrk@nallas-mac-Book-Air ~ % ./14
=====
PROCESS ABSTRACTION DEMO
=====

Parent Process Information
-----
Parent PID : 1986
Parent PPID : 1974
Process State: Running

=====
PARENT PROCESS
=====
Parent PID : 1986
Child PID : 1987
Parent PPID : 1974
Process State: Running

Parent is waiting for child...
Process State: Waiting
=====
CHILD PROCESS
=====
Child PID : 1987
Child PPID : 1986
Process State: Running

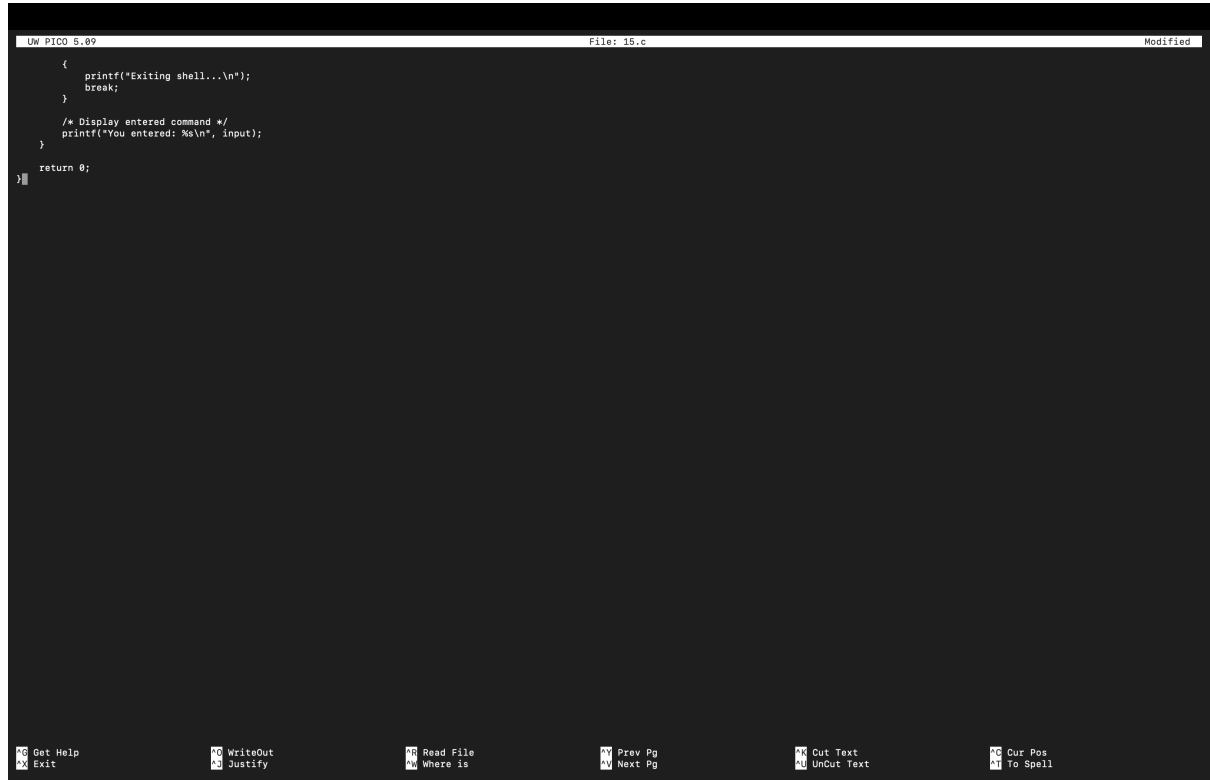
Child is going to execute 'ls' using exec().

total 3424
-rw----- 1 nrk staff 1447 26 Aug 12:06 1.c.save
-rwxr-xr-x 1 nrk staff 33472 29 Aug 08:36 11
-rw-r--r-- 1 nrk staff 802 29 Aug 08:35 11.c
-rwxr-xr-x 1 nrk staff 33608 29 Aug 08:47 13
-rw-r--r-- 1 nrk staff 1479 29 Aug 08:47 13.C
-rwxr-xr-x 1 nrk staff 33728 29 Aug 08:51 14
-rw-r--r-- 1 nrk staff 2349 29 Aug 08:51 14.c
-rwxr-xr-x 1 nrk staff 33804 26 Aug 12:11 2
-rw-r--r-- 1 nrk staff 1990 26 Aug 12:11 2.c
-rwxr-xr-x 1 nrk staff 33936 26 Aug 12:15 3
-rw-r--r-- 1 nrk staff 1142 26 Aug 12:14 3.c
-rwxr-xr-x 1 nrk staff 33768 26 Aug 12:28 4
-rw-r--r-- 1 nrk staff 1517 26 Aug 12:26 4.c
-rwxr-xr-x 1 nrk staff 33968 25 Aug 12:04 444
-rw-r--r-- 1 nrk staff 1202 25 Aug 12:03 444.c
-rwxr-xr-x 1 nrk staff 33768 26 Aug 12:28 5
-rw-r--r-- 1 nrk staff 1018 26 Aug 12:28 5.c
-rw----- 1 nrk staff 1394 26 Aug 11:57 a.c.save
drwxr-xr-x@ 4 nrk staff 128 16 Jul 17:12 airline-lost-found
-rw-r--r-- 1 nrk staff 1279292 24 May 19:31 ancient_dress.fbx
drwx-----@ 4 nrk staff 128 22 Mar 2025 Applications
-rw----- 1 nrk staff 1393 26 Aug 11:58 b.c.save
drwxr-xr-x@ 7 nrk staff 224 9 Jul 15:21 blenderkit_data
-rw-r--r--@ 1 nrk staff 4836 4 Jun 08:22 class AVLTree (.java
-rw-r--r--@ 1 nrk staff 1035 4 Jun 10:41 class RankNode (.java
drwxrwxrwx 2 nrk staff 64 22 Jun 13:04 DaVinci Resolve Media
-rw-r--r--@ 1 nrk staff 7131 29 Nov 2025 ddca.circ
drwx-----+ 4 nrk staff 128 12 Aug 08:37 Desktop
drwx-----+ 34 nrk staff 1088 29 Aug 08:48 Documents
drwx-----@ 97 nrk staff 3104 29 Aug 08:44 Downloads
drwxr-xr-x@ 3 nrk staff 96 15 Jun 10:35 EXAMINATION SEATING ARRANGEMENT
```

SESSION-2

Task 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

Code:

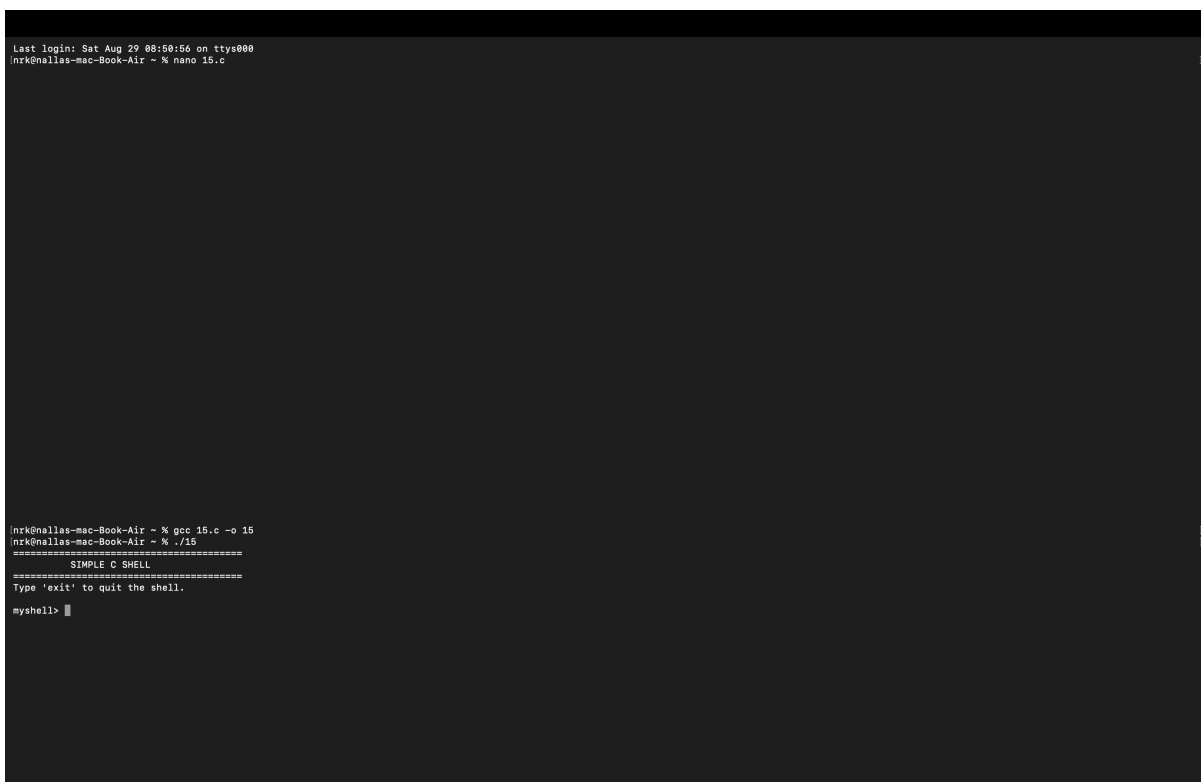


```
UW PICO 5.09 File: 15.c Modified
{
    printf("Exiting shell...\n");
    break;
}

/* Display entered command */
printf("You entered: %s\n", input);
}

return 0;
}
```

Output:

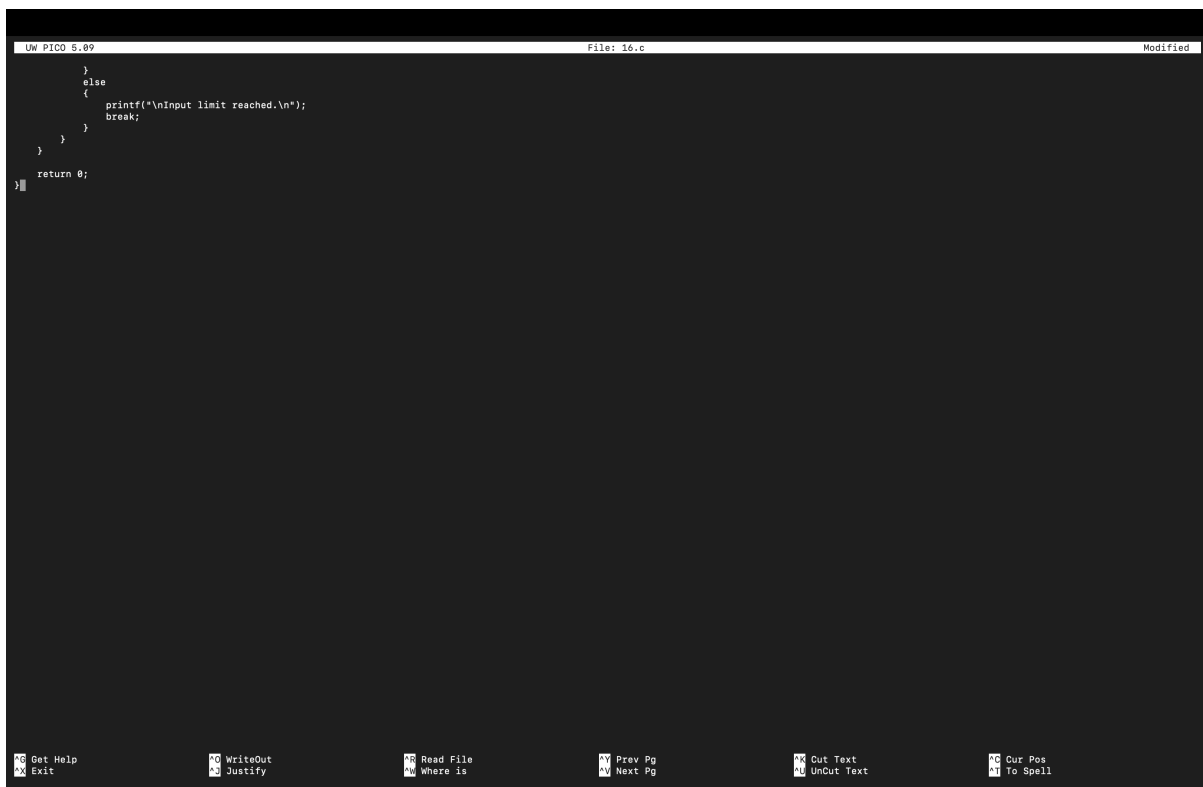


```
Last login: Sat Aug 29 08:50:56 on tty000
nrk@nallas-mac-Book-Air ~ % nano 15.c

nrk@nallas-mac-Book-Air ~ % gcc 15.c -o 15
nrk@nallas-mac-Book-Air ~ % ./15
=====
SIMPLE C SHELL
=====
Type 'exit' to quit the shell.
myshell> 
```

Task 2: To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction.

CODE:



The screenshot shows a code editor window with a dark background. The title bar at the top reads "UW PICO 5.09", "File: 16.c", and "Modified". The code is written in C and is as follows:

```
    }
    else
    {
        printf("\nInput limit reached.\n");
        break;
    }
}
return 0;
}
```

At the bottom of the editor, there is a status bar with several icons and labels: "Get Help", "Exit", "WriteOut", "Justify", "Read File", "Where is", "Prev Pg", "Next Pg", "Cut Text", "UnCut Text", "Cur Pos", and "To Spell".

```
Last login: Sat Aug 29 08:59:57 on ttys000
nrtk@nallas-mac-Book-Air ~ % nano 16.c
```

```
nrtk@nallas-mac-Book-Air ~ % gcc 16.c -o 16
nrtk@nallas-mac-Book-Air ~ % ./16
=====
      KEYBOARD INPUT HANDLING
=====
Type a command and press ENTER.
Use BACKSPACE to delete characters.
Type 'exit' to quit.

myshell> █
```